

ON THE CALCULATION OF THE FREQUENCY-DEPENDENT DIELECTRIC CONSTANT IN COMPUTER SIMULATIONS

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An equation is derived which relates, for all boundary conditions currently used, the dipole moment fluctuations observed in computer simulations of polar systems to the frequency-dependent dielectric constant. Provided that the modified dipolar interactions do not explicitly depend upon time, the resulting formula is of a generalized Glarum type.

1. Introduction

Relating the dielectric properties of matter — in fact, macroscopic properties in general — to its microscopic basis is the task of statistical mechanics. Thus, given a model system and a model interaction, statistical mechanics is expected to provide a prescription by means of which we should be able to calculate the system's dielectric constant, for instance, from first principles.

In the case of the static dielectric constant ϵ_0 , this relationship is due to Kirkwood [1] who has shown that, at temperature T , the equilibrium fluctuations of the total dipole moment M of a spherical volume (subsystem) V , cut from a very large sample of non-polarizable particles, are related to ϵ_0 by

$$\frac{4}{3}\pi\langle M^2 \rangle / 3Vk_B T = (2\epsilon_0 + 1)(\epsilon_0 - 1) / 9\epsilon_0. \quad (1)$$

The generalization of (1) to frequencies $\omega > 0$ has been given by Fatuzzo and Mason [2]:

$$\mathcal{L}_{i\omega}[-\dot{\Phi}] = \frac{[\epsilon(\omega) - 1][2\epsilon(\omega) + 1]\epsilon_0}{(\epsilon_0 - 1)(2\epsilon_0 + 1)\epsilon(\omega)}. \quad (2)$$

Here $\mathcal{L}_{i\omega}$ is the Fourier–Laplace transform and $\Phi(t) = \langle M(0) \cdot M(t) \rangle / \langle M^2 \rangle$ is the normalized autocor-

relation function of the total dipole moment of V . $\epsilon(\omega)$ is the frequency-dependent dielectric constant and $\epsilon_0 = \epsilon(0)$.

The assumption that both V and the rest of the system can be characterized by the same frequency-dependent dielectric constant is an essential point in the derivation of (2). Prior to Fatuzzo and Mason, Glarum [3] has published a relation differing from (2),

$$\mathcal{L}_{i\omega}[-\dot{\Phi}] = \frac{[\epsilon(\omega) - 1]3\epsilon_0}{(\epsilon_0 - 1)[\epsilon(\omega) + 2\epsilon_0]}. \quad (3)$$

It was Titulaer and Deutch [4] who finally showed that the two formulas simply correspond to different boundary conditions, with Glarum's formula applying to a spherical volume embedded in a dielectric continuum with frequency-independent dielectric constant ϵ_0 .

Finally, we have, for an isolated spherical sample in vacuo,

$$\frac{4}{3}\pi\langle M^2 \rangle / 3Vk_B T = (\epsilon_0 + 1) / (\epsilon_0 + 2) \quad (4)$$

for the static dielectric constant [5], and

$$\mathcal{L}_{i\omega}[-\dot{\Phi}] = \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \frac{\epsilon_0 + 2}{\epsilon_0 - 1} \quad (5)$$

for the frequency-dependent dielectric constant [3].

Now, the concept associated with the dielectric constant is that of a quantity depending only on the substance under consideration, no matter what the particular boundary conditions may happen to be. Therefore, in order for (1)–(5) to be mutually consistent, i.e. to correspond to the same dielectric constant $\epsilon(\omega)$, $\langle M^2 \rangle$ as well as $\Phi(t)$ have to respond to any change of boundary conditions in a characteristic way. From the point of view of calculating the dielectric constant in computer simulations of model systems, it thus proves essential to know whether eqs. (1)–(5) are at all applicable to the geometry commonly used in those simulations (cubic shape of V , toroidal boundary conditions, modified dipolar interactions), and how they may be generalized.

As far as the static dielectric constant is concerned, this problem has been solved, for very general boundary conditions, in two previous papers [6,7], where Kirkwood's result (1) as well as (4) could be obtained as special cases of a more general formula. In this article we apply our method to the case of the frequency-dependent dielectric constant. In particular, it will be shown that the Fatuzzo–Mason relation (2) cannot be obtained from the general formula for the frequency-dependent case, which is of the Glarum type and even contains (3) as a special case. Having been rehabilitated in this way, Glarum's result has now, in computer simulations, finally found a field of application, although not the one originally intended.

2. Theory

In the following we consider an isotropic system of N non-polarizable particles with molecular dipole moment μ , enclosed, at temperature T , in a cubic volume V to which toroidal boundary conditions are applied. Although V is cube-shaped in virtually all simulations, our results may be shown also to be valid for the more general class of basic cells (including, for example, the truncated octahedron and the rhombic d-decahedron) for which x , y and z are equivalent directions.

2.1. Linear response theory

Let $M(t) = \sum_{i=1}^N \mu_i(t)$ be the total dipole mo-

ment of the system at time t . According to linear response theory [8], the expectation value of M in the presence of a time-dependent but spatially homogeneous external field $E_0(t)$ is given by

$$\langle M(t) \rangle_{E_0} = -\frac{1}{3k_B T} \int_0^\infty dt' E_0(t-t') d\langle M(0) \cdot M(t') \rangle / dt' . \quad (6)$$

Since we may regard V as being isolated as well as finite (interactions extending beyond V , like those required by lattice summation techniques, may be rewritten as effective interactions within V), the field appearing on the rhs of (6) is always the external field E_0 itself rather than the cavity field typical of embedded geometries.

Eq. (6) is of the convolution type. Therefore, the Fourier components of the dielectric polarization $P(t) = M(t)/V$ and the external field E_0 are related by

$$\tilde{P}(\omega) = (\langle M^2 \rangle / 3Vk_B T) \mathcal{L}_{i\omega}[-\dot{\Phi}] \tilde{E}_0(\omega) , \quad (7)$$

where $\mathcal{L}_{i\omega}$ denotes the Fourier–Laplace transform

$$\mathcal{L}_{i\omega}[-\dot{\Phi}] = \int_0^\infty dt \exp(-i\omega t) (-d\Phi/dt) \quad (8)$$

and $\Phi(t)$ is the normalized autocorrelation function of $M(t)$.

2.2. Modified dielectric theory

In order to yield an expression for the frequency-dependent dielectric constant, (7) has to be combined with the corresponding result of continuum theory. A convenient starting point is the fundamental equation of continuum theory,

$$E(r, t) = E_0(r, t) + \int dr' \mathbf{T}(r-r') \cdot P(r', t) , \quad (9)$$

which states that, in the presence of polarizable matter, the resulting electric field (Maxwell field) E is given by the external field E_0 plus contributions due to the macroscopic polarization P of the system. In (9) $\mathbf{T}$ is the dipole–dipole tensor

$$T_{ab}(r) = \nabla_a \nabla_b (1/r) = 3r_a r_b / r^5 - \delta_{ab} / r^3 \quad (10)$$

and retardation effects have been neglected.

From the point of view of understanding the influence of boundary conditions on computer simula-

tions of polar systems, it is essential to realize that (9) is simply the macroscopic analogue of

$$E_i(t) = E_0(r_i(t), t) + \sum_{j=1}^N \mathbf{T}_{\text{mod}}(r_i(t) - r_j(t)) \cdot \boldsymbol{\mu}_j(t) \quad (11)$$

by means of which the total electric field E_i , acting on particle i , is calculated in simulations. However, despite the analogy, the microscopic version differs from (9) in that toroidal boundary conditions are implicitly assumed to apply in (11); that is, a particle j is replaced by its nearest image j' whenever $|r_i(t) - r_j(t)|$ is greater than half the cell length. Also, we have used the special symbol T_{mod} to make it clear that, in computer simulations, dipolar interactions usually are not given by the original T tensor (10), but by one of the modifications discussed in section 3 instead.

In a computer-adapted version of (9) both facts will have to be taken into account. For toroidal boundary conditions over a finite volume, this can be done automatically by switching over to spatial or — since we will need frequency components later on — double Fourier transforms [6]. Therefore, the modified version of (9) now reads

$$\hat{E}(k, \omega) = \hat{E}_0(k, \omega) - \frac{4}{3}\pi \hat{P}(k, \omega) + \tilde{\mathbf{T}}_{\text{mod}}(k) \cdot \hat{P}(k, \omega), \quad (12)$$

where

$$\hat{P}(k, \omega) = \int_{-\infty}^{\infty} dt \exp(-i\omega t) \int_V dr \exp(-ik \cdot r) P(r, t) \quad (13)$$

and $\hat{E}$ and $\hat{E}_0$ are defined analogously. Note, however, that T_{mod} does not depend upon time, so that only the spatial Fourier transform

$$\tilde{\mathbf{T}}_{\text{mod}}(k) = \int_V dr \exp(-ik \cdot r) \mathbf{T}_{\text{mod}}(r) \quad (14)$$

enters into (12). In (14) the origin is to be excluded from the integral, the singularity $\lim_{r \rightarrow 0} \mathbf{T}_{\text{mod}}(r) = -\frac{4}{3}\pi \delta(r) \mathbf{T}$ having been explicitly accounted for in (12).

In order to compare (12) to (7), the linear response result for homogeneous perturbations, we need the $k \rightarrow 0$ limit of the former. Writing $\hat{P}(k=0, \omega) = \tilde{P}(\omega)$ and so on, and eliminating $\tilde{E}$ by means of the constitutive relation

$$\tilde{P}(\omega) = \frac{\epsilon(\omega) - 1}{4\pi} \tilde{E}(\omega), \quad (15)$$

we obtain, as the macroscopic analogue of (7),

$$\tilde{P}(\omega) = \frac{3}{4\pi} \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \left(1 - \frac{3}{4\pi} \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \tilde{\mathbf{T}}_{\text{mod}}(0) \right)^{-1} \tilde{E}_0(\omega) \quad (16)$$

(here we have already used $\tilde{\mathbf{T}}_{\text{mod}}(0) = \tilde{\mathbf{T}}_{\text{mod}}(0)$ for all modified $\mathbf{T}$ tensors discussed in section 3). Combination of (7) and (9) now yields

$$\frac{4}{3}\pi \langle M^2 \rangle / 3 V k_B T \mathcal{L}_{i\omega}[-\Phi] = \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \left(1 - \frac{3}{4\pi} \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \tilde{\mathbf{T}}_{\text{mod}}(0) \right)^{-1}. \quad (17)$$

In the limit $\omega \rightarrow 0$ we have $\mathcal{L}_{i\omega}[-\Phi] = 1 - i\omega \mathcal{L}_{i\omega}[\Phi] \rightarrow 1$, so that, for the static dielectric constant ϵ_0 , (17) becomes

$$\frac{4}{3}\pi \langle M^2 \rangle / 3 V k_B T = \frac{\epsilon_0 - 1}{\epsilon_0 + 2} \left(1 - \frac{3}{4\pi} \frac{\epsilon_0 - 1}{\epsilon_0 + 2} \tilde{\mathbf{T}}_{\text{mod}}(0) \right)^{-1}, \quad (18)$$

which is the result familiar from [6,7]. Combining (17) and (18) we finally obtain

$$\mathcal{L}_{i\omega}[-\Phi] = \frac{\epsilon(\omega) - 1}{\epsilon(\omega) + 2} \frac{\epsilon_0 + 2}{\epsilon_0 - 1} \times \frac{1 - (3/4\pi)[(\epsilon_0 - 1)/(\epsilon_0 + 2)] \tilde{\mathbf{T}}_{\text{mod}}(0)}{1 - (3/4\pi)[\{\epsilon(\omega) - 1\}/\{\epsilon(\omega) + 2\}] \tilde{\mathbf{T}}_{\text{mod}}(0)}. \quad (19)$$

This is our central result. We are now in a position to calculate, for any type of boundary conditions, the frequency-dependent dielectric constant of a model system from the equilibrium autocorrelation function of its total dipole moment — provided the modified interaction $u_{\text{DD}}(i, j) = -\boldsymbol{\mu}_i \cdot \mathbf{T}_{\text{mod}}(i, j) \cdot \boldsymbol{\mu}_j$ does not explicitly depend upon time.

3. Specific forms of interaction T_{mod}

The modified interactions most frequently used in the literature may be roughly grouped into two classes, depending on the number of particles in the system which are allowed to interact. If only pairs of particles separated by a distance $r_{ij} < r_c$ are allowed to interact, we may write [‡]

[‡] The influence of smoothing (tapering) functions is discussed in ref. [6].

$$\mathbf{T}_{\text{mod}}(r) = \mathbf{T}(r) + \frac{1}{r_c^3} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \mathbf{I}, \text{ if } r < r_c, \\ = 0, \text{ otherwise,} \quad (20)$$

while, if there are no such restrictions, $\mathbf{T}_{\text{mod}}$ is of the general form

$$\mathbf{T}_{\text{mod}}(r) = \mathbf{T}_{\Sigma}(r) + \frac{4\pi}{3V} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \mathbf{I}, \quad (21)$$

with

$$\mathbf{T}_{\Sigma}(r) = \mathbf{T}(r) + \sum_{n \neq (0,0,0)} \mathbf{T}(r + nL). \quad (22)$$

In these equations, $\mathbf{T}$ is the unmodified dipole-dipole interaction (19), L is the length of the basic cell, i.e. $V = L^3$, and the sum in (22) is over all integer vectors except $n = (0,0,0)$.

Among the special cases covered by eq. (20) are: spherical truncation of $\mathbf{T}$ (so-called spherical-cutoff geometry), for $\epsilon_{\text{RF}} = 1$; and spherical reaction field geometry, for $\epsilon_{\text{RF}} > 1$ (preferably $\epsilon_{\text{RF}} \approx \epsilon_0$) [9].

Eq. (21), on the other hand, is seen to reduce to minimum-image geometry in the special case where $\epsilon_{\text{RF}} = 1$ and the sum is omitted from (22), so that the interaction is given by the original $\mathbf{T}$ tensor. In the general case, however, i.e. including the sum in $\mathbf{T}_{\Sigma}$, (21) represents the form of interaction typically used by lattice summation techniques, which are the type of boundary conditions most frequently met in the literature. $\mathbf{T}_{\Sigma}$ has been introduced as a special symbol because, in our treatment, we need to rewrite interactions between particles in *different* cells as *effective* interactions *within* the basic cell. How the lattice sum is actually realized in a simulation, whether in the form given by Ewald and Kornfeld [10], Bertaut [11] or Ladd [12], is of secondary importance as long as it is equivalent to a summation over spherical shells of lattice points (image cells) in (22). It has been shown that $\tilde{\mathbf{T}}_{\Sigma}(0) = 0$ in that case [7] (the term offending this rule in the Ewald method can be extracted from the sum and may then be interpreted as a formal reaction field given by the second term in (21)).

According to (14), the $k = 0$ component of $\tilde{\mathbf{T}}_{\text{mod}}$ is given by the integral of $\mathbf{T}_{\text{mod}}$ over the entire volume V . However, because of the cutoff $\mathbf{T}_{\text{mod}}(r) = 0$ for $r > r_c$, (20) reduces to an integral over a sphere of radius r_c , integrated over which the unmodified $\mathbf{T}$

tensor, owing to its symmetry, is known to vanish, i.e. $\tilde{\mathbf{T}}(0) = 0$. Therefore, in both cases, (20) and (21), $\tilde{\mathbf{T}}_{\text{mod}}(0)$ is solely determined by the reaction field term which is proportional to the unit tensor. Because of the different range of interaction (and integration) in (20) and (21), both reaction field terms even make the same contribution, so that

$$\tilde{\mathbf{T}}_{\text{mod}}(0) = \frac{4}{3}\pi \frac{2(\epsilon_{\text{RF}} - 1)}{(2\epsilon_{\text{RF}} + 1)} \mathbf{I} \quad (23)$$

in either case.

Thus, a single number, namely ϵ_{RF} , is sufficient to characterize the whole spectrum of boundary conditions, at least as far as their influence on the global (i.e. zero wave vector) dielectric properties is concerned. Although we will usually speak of ϵ_{RF} as of the external static dielectric constant this does not necessarily imply that there is a "real" (or even imaginary) dielectric continuum in the system being simulated. ϵ_{RF} may also have been introduced on purely formal grounds, as in the case of the Ewald sum [7]. Therefore, the concept of an external dielectric constant should simply be understood as a convenient and compact way of classifying boundary conditions, and it will be used in that sense in our discussion in section 4.

Inserting (23) in (18) and (19) we obtain, no matter what $\mathbf{T}_{\text{mod}}$ looks like,

$$\mathcal{P}_{i\omega}[-\dot{\Phi}] = \frac{\epsilon(\omega) - 1}{\epsilon_0 - 1} \frac{2\epsilon_{\text{RF}} + \epsilon_0}{2\epsilon_{\text{RF}} + \epsilon(\omega)}, \quad (24)$$

with ϵ_0 given by

$$\frac{4}{3}\pi \langle M^2 \rangle / 3V k_B T = \frac{\epsilon_0 - 1}{3} \frac{2\epsilon_{\text{RF}} + 1}{2\epsilon_{\text{RF}} + \epsilon_0}. \quad (25)$$

In contrast to the formulae given in section 1, (24) and (25) now hold for arbitrary isotropic samples (cube, truncated octahedron, rhombic dodecahedron, etc.) with periodic boundary conditions. By specifying appropriate values for ϵ_{RF} , we are even able to recover some of the formulae of section 1, which are thus found to be valid for non-spherical geometry also.

For example, setting $\epsilon_{\text{RF}} = 1$ in (24) and (25), one immediately obtains (4) and (5), which therefore apply not only to spherical samples but also to spherical cutoff and minimum image geometry as well as lattice sums without explicit reaction field [cf. the discussion following eqs. (20)–(22)].

In the limit $\epsilon_{\text{RF}} \rightarrow \infty$, on the other hand, we ob-

tain

$$\rho_{i\omega}[-\dot{\Phi}] = [\epsilon(\omega) - 1]/(\epsilon_0 - 1) \quad (26)$$

and

$$\frac{4}{3}\pi\langle M^2 \rangle / 3Vk_B T = \frac{1}{3}(\epsilon_0 - 1), \quad (27)$$

which are the equations appropriate for spherical reaction field geometry (with $\epsilon_{RF} = \infty$) as well as Ewald simulations with so-called conducting boundary conditions (cf. refs. [13,14] but also ref. [7]).

In the special case $\epsilon_{RF} = \epsilon_0$, eq. (25) reduces to the Kirkwood formula (1) [6]. Surprisingly – or perhaps not – the analogous result for the frequency-dependent dielectric constant, following from (24), is not the formula of Fatuzzo and Mason but that of Glarum. In fact, given the structure of eq. (19), i.e. a T_{mod} independent of time, there is no way the general relation (24) can lead to the Fatuzzo–Mason result, no matter what we choose for ϵ_{RF} . However, this is not intended to be a revival of the discussions concerning the validity of the Fatuzzo–Mason versus the Glarum relation. It simply means that the dielectric formulae appropriate for computer simulations involving dipolar interactions of the type (20)–(22) are different from those describing the dielectric relaxation of a spherical subsystem within a truly infinite sample.

4. Dependence of $\Phi(t)$ on boundary conditions

It seems reasonable to assume that, for a given substance, $\epsilon(\omega)$ does not depend on boundary conditions. Strictly speaking, this is just how the dielectric constant is defined. With $\Phi(t)$, however, things are entirely different. For, according to eq. (19) or (24), boundary conditions (i.e. the particular form of T_{mod}) should have a very strong and characteristic influence on $\Phi(t)$.

In order to simplify the discussion, we will assume that the system being simulated behaves like a Debye dielectric, whose frequency-dependent dielectric constant is given by [15]

$$[\epsilon(\omega) - 1]/(\epsilon_0 - 1) = 1/(1 + i\omega\tau_D). \quad (28)$$

Here τ_D is the Debye relaxation time, and we have set $\epsilon_\infty = 1$ for a system of non-polarizable particles.

Characterizing, as in section 3, the variety of

boundary conditions by a suitable external dielectric constant ϵ_{RF} , we may insert (28) into (24), obtaining

$$\Phi(t) = \exp(-t/\tau), \quad (29)$$

where

$$\tau = [(2\epsilon_{RF} + 1)/(2\epsilon_{RF} + \epsilon_0)] \tau_D. \quad (30)$$

Thus, for the Debye model, $\Phi(t)$ is a pure exponential whose relaxation time depends on ϵ_{RF} , i.e. on boundary conditions, in a characteristic way. Another interesting relation can be obtained by combining (30) with (25):

$$\tau/\tau_D = [4\pi/(\epsilon_0 - 1)] \langle M^2 \rangle / 3Vk_B T. \quad (31)$$

That is, in a Debye dielectric, a linear relationship holds between τ and $\langle M^2 \rangle$: relaxation times and dipole moment fluctuations are influenced by boundary conditions in exactly the same way.

In actual simulations, the following cases will be of particular interest

$$\tau = [3/(\epsilon_0 + 2)] \tau_D, \quad \text{if } \epsilon_{RF} = 1, \quad (32a)$$

$$= [(2\epsilon_0 + 1)/3\epsilon_0] \tau_D, \quad \text{if } \epsilon_{RF} = \epsilon_0, \quad (32b)$$

$$= \tau_D, \quad \text{if } \epsilon_{RF} = \infty. \quad (32c)$$

The decay functions corresponding to these relaxation times have been plotted in fig. 1, together with the Fatuzzo–Mason result

$$\Phi(t) = (2\epsilon_0 + 1)^{-1} [2\epsilon_0 \exp(-t/\tau_D) + \exp(-\epsilon_0 t/\tau_D)], \quad (33)$$

which, for a Debye dielectric, is a sum of two exponentials [16]. In fig. 1 it looks as if the Fatuzzo–Mason curve were interpolating between the Glarum result (32b) (at short times) and conducting boundary conditions (32c) (at long times). Actually, this equivalence at short times of Fatuzzo and Mason's result with that of Glarum is not a feature typical of the Debye model [in which case it follows from the series expansion of (29) and (33)], but holds quite generally: at short times, i.e. in the limit $\omega \rightarrow \infty$, we have $\epsilon(\omega) \approx 1$, so that (2) is almost identical with (3). For long times ($\omega \rightarrow 0$), however, embedded geometry and conducting boundary conditions will only be equivalent if ϵ_0 [and thus $\epsilon(\omega \approx 0)$] is sufficiently large [cf. eqs. (2) and (26)].

Since a great number of substances can be ade-

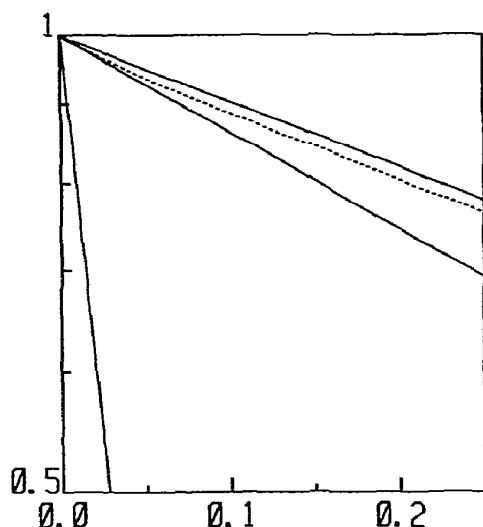


Fig. 1. $\Phi(t)$ for a Debye dielectric with $\epsilon_0 = 25$ and various boundary conditions. Solid lines (top to bottom): conducting boundary conditions ($\epsilon_{RF} = \infty$), reaction field geometry ($\epsilon_{RF} = \epsilon_0$), and spherical cutoff geometry ($\epsilon_{RF} = 1$). Dashed line: embedded sphere [Fatuzzo-Mason result, eq. (33)]. Time is in units of τ_D .

quately modeled by a Debye dielectric (most correlation functions do become exponentials sooner or later), the qualitative pattern visible in fig. 1 should be typical of the influence of boundary conditions on the behavior of simulated systems also. In particular, we expect $\Phi(t)$ to decay much more rapidly in simulations without reaction field (spherical cutoff with $\epsilon_{RF} = 1$, minimum image, and lattice sums without explicit reaction field) than in simulations

including a reaction field of one kind or another. Of course, this effect will be more conspicuous the higher the static dielectric constant of the model system happens to be. On the other hand, the correlation function obtained from a simulation with conducting boundary conditions ($\epsilon_{RF} = \infty$) will decay more slowly than the one corresponding to a reaction field simulation with $\epsilon_{RF} \approx \epsilon_0$.

To what extent the predictions of our modified dielectric theory are obeyed in actual simulations will be investigated in a subsequent publication.

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